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1. ★ ★ ★ ★ ★
★ ★
★ ★ $n = 5$
★ ★
★ ★ ★ ★ ★

intialization ; $i = 1$; $j = 1$

iteration-1

condition ; $i \le n$; $j \le n$

$i=1$ $1 \le 5$ ✓ ; $j = 1$; $1 \le 5$; ✓
$i=1$ $1 \le 5$ ✓ ; $j = 2$; $2 \le 5$; ✓
$i=1$ $1 \le 5$ ✓ ; $j = 3$; $3 \le 5$; ✓
$i=1$ $1 \le 5$ ✓ ; $j = 4$; $4 \le 5$; ✓
$i=1$ $1 \le 5$ ✓ ; $j = 5$; $5 \le 5$; ✓
$i=1$ $1 \le 5$ ✓ ; $j = 6$; $6 \le 5$; ✗

print ★ → if ($i = 1$ || $i = n$; $j = 1$; $j = n$) else (" ")

$i=1$ $j=1$ print → (★) ✓ $i = 1$; $j = 1$ ✓
$i=1$ $j=2$ print → (★) ✓ $i = 1$; $j = 2$ ✗
$i=1$ $j=3$ print → (★) ✓ $i = 1$; $j = 3$ ✗
$i=1$ $j=4$ print → (★) ✓ $i = 1$; $j = 4$ ✗
$i=1$ $j=5$ print → (★) ✓ $i = 1$; $j = 5$ ✗

iteration-2

$i=2$ $2 \le 5$ ✓ ; $j = 1$; $1 \le 5$ ✓
$i=2$ $2 \le 5$ ✓ ; $j = 2$; $2 \le 5$ ✓
$i=2$ $2 \le 5$ ✓ ; $j = 3$; $3 \le 5$ ✓
$i=2$ $2 \le 5$ ✓ ; $j = 4$; $4 \le 5$ ✓
$i=2$ $2 \le 5$ ✓ ; $j = 5$; $5 \le 5$ ✓
$i=2$ $2 \le 5$ ✓ ; $j = 6$; $6 \le 5$ ✗

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Print.

i = 1 ; j = 1 ; j = 1 ✓ print (★)
i = 2 ; j = 2 ; i = 1 ✗ || j = 1 ✗ || i = 5 ✗ || j = 5 ✗ print (" ")
i = 2 ; j = 3 ; i = 1 ✗ || j = 1 ✗ || i = 5 ✗ || j = 5 ✗ print (" ")
i = 2 ; j = 4 ; i = 1 ✗ || j = 1 ✗ || i = 5 ✗ || j = 5 ✗ print (" ")
i = 2 ; j = 5 ; i = 1 ✗ || j = 1 ✗ || j = 5 ✓ || i = 5 ✗ print (★)

Iteration 3 & 4 Same as (2)

Iteration 5
Condition:
i = 5 5 ≤ 5 ; j = 1 ; 1 ≤ 5 ✓ ★
i = 5 5 ≤ 5 ; j = 2 ; 2 ≤ 5 ✓
i = 5 5 ≤ 5 ; j = 3 ; 3 ≤ 5 ✓
i = 5 5 ≤ 5 ; j = 4 ; 4 ≤ 5 ✓
i = 5 5 ≤ 5 ; j = 5 ; 5 ≤ 5 ✓
i = 5 5 ≤ 5 ; j = 6 ; 6 ≤ 5 ✗

Print:
i = 5 j = 1 ; i = 1 ✗ || i = 5 ✓ || j = 1 ✓ || j = 5 ✗ print (★)
i = 5 j = 2 ; i = 1 ✗ || i = 5 ✓ || j = 1 ✗ || j = 5 ✗ print (★)
i = 5 j = 3 ; i = 1 ✗ || i = 5 ✓ || j = print (★)
i = 5 j = 4 ; i = 1 ✗ || i = 5 ✓ || j = print (★)
i = 5 j = 5 ; i = 1 ✗ || i = 5 ✓ || j = print (★)

iteration : 6

i = 6 ; 6 ≤ 5 ✗ — — — — —

```
*       *
  *   *
    *
  *   *
*       *
```

$n = 5$

intialization ; $i = 1$; $j = 1$;

condition ; $i \le n$; $j \le n$; $n = 5$

iteration = 1

$i=1$	$1 \le 5$ ✓ ;	$j=1$	$1 \le 5$	✓
$i=1$	$1 \le 5$ ✓ ;	$j=2$	$2 \le 5$	✓
$i=1$	$1 \le 5$ ✓ ;	$j=3$	$3 \le 5$	✓
$i=1$	$1 \le 5$ ✓ ;	$j=4$	$4 \le 5$	✓
$i=1$	$1 \le 5$ ✓ ;	$j=5$	$5 \le 5$	✓
$i=1$	$1 \le 5$ ✓ ;	$j=6$	$6 \le 5$	X

print * → if $(i=j)$ || $i+j = n+1$
else print (" ")

$i=1$	$j=1$	$i=j$ ✓ || $i+j \ne 6$ X →	print ("*")
$i=1$	$j=2$	$i=j$ X || $i+j \ne 6$ X →	print (" ")
$i=1$	$j=3$	$i=j$ X || $i+j \ne 6$ X →	print (" ")
$i=1$	$j=4$	$i=j$ X || $i+j \ne 6$ X	print (" ")
$i=1$	$j=5$	$i=j$ X || $i+j = 6$ ✓	print (" ")

iteration = 2

$i=2$	$2 \le 5$	$j=1$;	$1 \le 5$	✓
$i=2$	$2 \le 5$	$j=2$;	$2 \le 5$	✓
$i=2$	$2 \le 5$	$j=3$;	$3 \le 5$	✓
$i=2$	$2 \le 5$	$j=4$;	$4 \le 5$	✓
$i=2$	$2 \le 5$	$j=5$;	$5 \le 5$	✓
$i=2$	$2 \le 5$	$j=6$;	$6 \le 5$	X

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Similar for iteration 3 4 5

Print :

i=2	j=1 ;	i=j ✗ || i+j=6 ✗	print(" ");
i=2	j=2 ;	i=j ✓ || i+j=6 ✗	Print (★) ;
i=2	j=3 ;	i=j ✗ || i+j=6 ✗	Print(" ");
i=2	j=4 ;	i=j ✗ || i+j=6 ✓	Print (★);
i=2	j=5 ;	i=j ✗ || i+j=6 ✗	print(" ");

Similar for iteration 3 4 5

When i = 6

i ≤ 6 ✗

Q.3 ★

★ ★

★ ★ ★ $n = 5$.

★ ★ ★ ★

★ ★ ★ ★ ★

intialization: $i = 1$ $j = 1$.

Condition $i <= n$; $j = i$ $n = 5$

★ iteration 1

$i = 1$	$1 \le 5$ ✓	$j = 1$	$1 \le 1$ ✓	print(★)
$i = 2$	$2 \le 5$ ✓	$j = 2$	$2 \le 1$ X	—

★ iteration 2

$i = 2$	$2 \le 5$ ✓	$j = 1$	$1 \le 2$ ✓	Print(★)
$i = 2$	$2 \le 5$ ✓	$j = 2$	$2 \le 2$ ✓	print(★)
$i = 2$	$2 \le 5$ ✓	$j = 3$	$3 \le 2$ X	—

★ iteration 3

$i = 3$	$3 \le 5$ ✓	$j = 1$	$1 \le 3$ ✓	print(★)
$i = 3$	$3 \le 5$ ✓	$j = 2$	$2 \le 3$ ✓	Print(★)
$i = 3$	$3 \le 5$ ✓	$j = 3$	$3 \le 3$ ✓	Print(★)
$i = 3$	$3 \le 5$ ✓	$j = 4$	$4 \le 3$ X	—

★ iteration 4.

$i = 4$	$4 \le 5$ ✓	$j = 1$	$1 \le 4$ ✓	Print(★)
$i = 4$	$4 \le 5$ ✓	$j = 2$	$2 \le 4$ ✓	print(★)
$i = 4$	$4 \le 5$ ✓	$j = 3$	$3 \le 4$ ✓	print(★)
$i = 4$	$4 \le 5$ ✓	$j = 4$	$4 \le 4$ ✓	print(★)
$i = 4$	$4 \le 5$ ✓	$j = 5$	$5 \le 4$ X	—

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★ iteration 5 :-

$i=5$	$5 \le 5$ ✓	$j=1$	$1 \le 5$ ✓	print(★)
$i=5$	$5 \le 5$ ✓	$j=2$	$2 \le 5$ ✓	print(★)
$i=5$	$5 \le 5$ ✓	$j=3$	$3 \le 5$ ✓	print(★)
$i=5$	$5 \le 5$ ✓	$j=4$	$4 \le 5$ ✓	print(★)
$i=5$	$5 \le 5$ ✓	$j=5$	$5 \le 5$ ✓	print(★)
$i=5$	$5 \le 5$ ✓	$j=6$	$6 \le 5$ X	—

★ $i=6$

$i=6$ — $6 \le 5$ X — — — —

```
* * * * *
* * * *
* * *        n = 5
* *
*
```

initialization = $i = n$ $j = 1$.

Condition : $i > 0$ $j \le i$

iteration 1

$i = 5$ $5 > 0$ ✓ ; $j = 1 \le 5$ ✓ print (*)
$i = 5$ $5 > 0$ ✓ ; $j = 2 \le 5$ ✓ print (*)
$i = 5$ $5 > 0$ ✓ ; $j = 3 \le 5$ ✓ print (*)
$i = 5$ $5 > 0$ ✓ ; $j = 4 \le 5$ ✓ print (*)
$i = 5$ $5 > 0$ ✓ ; $j = 5 \le 5$ ✓ print (*)
$i = 5$ $5 > 0$ ✓ ; $j = 6 \le 5$ X

iteration 2

$i = 4$ $4 > 0$ ✓ ; $j = 1 \le 4$ ✓ Print (*)
$i = 4$ $4 > 0$ ✓ ; $j = 2 \le 4$ ✓ print (*)
$i = 4$ $4 > 0$ ✓ ; $j = 3 \le 4$ ✓ print (*)
$i = 4$ $4 > 0$ ✓ ; $j = 4 \le 4$ ✓ print (*)
$i = 4$ $4 > 0$ ✓ ; $j = 5 \le 4$ X ~~print~~ -

iteration 3

$i = 3$ $3 > 0$ ✓ ; $j = 1$ ✓ Print (*)
$i = 3$ $3 > 0$ ✓ ; $j = 2$ ✓ print (*)
$i = 3$ $3 > 0$ ✓ ; $j = 3$ ✓ Print (*)
$i = 3$ $3 > 0$ ✓ ; $j = 4$ X ~~print~~ -

iteration 4

$i = 2$ $2 > 0$ ✓ ; $j = 1 \le 2$ ✓ Print (*)
$i = 2$ $2 > 0$ ✓ ; $j = 2 \le 2$ ✓ print (*)
$i = 2$ $2 > 0$ ✓ ; $j = 3 \le 2$ X

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iteration 5.

$i = 1$; $1 > 0$; $j = 1$ - $1 \leq 1$ ✓ print ("*")

$i = 1$; $1 > 0$; $j = 2$ - $2 \leq 1$ ✗

iteration : 6

$i = 0$; $0 > 0$ ✗ — — — —

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★ # ★
★ # n = 3
★ # ★

intialization i = 1 ; j = 1 ; n = 3

★ Condition : $i \le n$ $j \le n$

i = 1 1 ≤ 5 ✓ ; j = 1 ; j ≤ 3 ✓
i = 1 1 ≤ 5 ✓ ; j = 2 ; j ≤ 3 ✓
i = 1 1 ≤ 5 ✓ ; j = 3 ; j ≤ 3 ✓
i = 1 1 ≤ 5 ✓ ; j = 4 ; j ≤ 3 X

★ print if (i % 2 != 0) && (j % 2 = 0) → print (#)
· else if (i % 2 == 0 && j % 2 != 0) → print (#)
else (★)

i = 1 j = 1 (i % 2 != 0) ✓ && (j % 2 = 0) X → X
(i % 2 == 0) X && (j % 2 != 0) ✓ → X
else Statement ✓ → print (★)

i = 1 j = 2 (i % 2 != 0) ✓ && (j % 2 = 0) ✓ → print (#)
(i % 2 == 0) X && (j % 2 != 0) X → X
else Statement → X

i = 1 j = 3 (i % 2 != 0) ✓ && (j % 2 = 0) X → X
(i % 2 == 0) X && (j % 2 != 0) ✓ → X
else Statement → print (★)
⋮ ⋮ ⋮ ⋮

Similar for i = 2 , i = 3